Predictive Values and Prevalence

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Introduction

Sensitivity and specificity are properties of a diagnostic test, not its clinical utility. **Positive predictive value (PPV)** – the probability that a positive test indicates disease – and **negative predictive value (NPV)** depend on disease prevalence. The same test can have very high PPV in a high-prevalence setting and very low PPV in a screening setting.

Prerequisites

Sensitivity and specificity; Bayes' theorem.

Theory

$$\text{PPV} = \frac{\text{Sens} \cdot \text{Prev}}{\text{Sens} \cdot \text{Prev} + (1 - \text{Spec}) \cdot (1 - \text{Prev})}.$$

$$\text{NPV} = \frac{\text{Spec} \cdot (1 - \text{Prev})}{(1 - \text{Sens}) \cdot \text{Prev} + \text{Spec} \cdot (1 - \text{Prev})}.$$

For a fixed test, PPV rises with prevalence and NPV falls. A high-specificity test is essential for screening in low-prevalence populations.

Assumptions

Test characteristics (Sens, Spec) generalise to the target population; prevalence is correctly estimated.

R Implementation

```r
library(epiR)

# 2x2 table from a diagnostic study
tab <- as.table(matrix(c(90, 10,     # TP, FP
                          20, 880),   # FN, TN
                        nrow = 2, byrow = FALSE,
                        dimnames = list(Test = c("+", "-"),
                                        Disease = c("yes", "no"))))

epi.tests(tab, conf.level = 0.95)

# What happens when prevalence is only 1%?
sens <- 0.82; spec <- 0.99
for (p in c(0.5, 0.2, 0.05, 0.01)) {
  ppv <- sens * p / (sens * p + (1 - spec) * (1 - p))
  npv <- spec * (1 - p) / ((1 - sens) * p + spec * (1 - p))
  cat(sprintf("Prev=%.2f  PPV=%.3f  NPV=%.3f\n", p, ppv, npv))
}
```

Output & Results

Test statistics including PPV/NPV in the sample; manual computation shows PPV dropping sharply as prevalence falls, from 0.98 at 50 % prevalence to 0.45 at 1 %.

Interpretation

"In a population with 1 % prevalence, a positive test (sens 82 %, spec 99 %) has a PPV of only 0.45; most positives are false. For screening, test specificity drives PPV far more than sensitivity does."

Practical Tips

- Always report PPV and NPV at the **target** population's prevalence, not the study sample's.
- For low-prevalence settings, confirmatory testing after a positive screen is usually essential.
- Likelihood ratios avoid dependence on prevalence and combine multiplicatively with prior odds.
- Bayes' post-test probability: $P(D \mid +) = \mathrm{LR}(+) \cdot \mathrm{Prev}/(1 + \mathrm{LR}(+) \cdot \mathrm{Prev})$ using prior odds.
- Decision curves (Vickers-Elkin) integrate PPV / NPV at different thresholds into a utility measure.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Diagnostic testing: Se, Sp, PPV, NPV, LR
- Kappa, ICC, Bland–Altman
- Biomarker statistics (Youden, NRI, decision curves)
- TRIPOD-AI, fairness auditing, reproducibility at scale